

Yuchen Liu

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AI systems engineer building RL training infrastructure, search/data pipelines, and research-grade open-source tools.

EDUCATION

M.S. Computer Science	University of Pennsylvania, Philadelphia	GPA: 3.80/4.00	May 2025
B.S. Applied Economics	Pennsylvania State University, State College	GPA: 3.86/4.00	May 2023

WORK EXPERIENCE

Software Engineer, Search Infrastructure | DoorDash Aug 2025 - Present | Seattle, WA

- Built Notus, a next-generation search ingestion platform using Iceberg, Kafka, and Spark to decouple document ingestion from serving, enable configuration-driven indexing, and unblock schema evolution for hybrid search.
- Built Vela, Argo's hybrid retrieval platform, enabling Lucene-native hybrid search and federated retrieval with vector backends such as Milvus behind a single API and abstracting backend complexity from customers.
- Migrated 30+ production stacks to modular Jsonnet, eliminating roughly 56,000 lines of duplicate configuration and reducing config verification time by 90%.
- Stabilized production indexers by migrating workloads to 500GB EBS storage and adding pre-deployment validators, preventing capacity-related outages after deployment.
- Led RCA work for revenue-impacting incidents with \$21K at risk, implementing Kyverno policies and automated guardrails to prevent traffic loss and human error.

Software Engineering Intern | Penn Medicine TissueLab Oct 2024 - May 2025 | Philadelphia, PA

- Developed a Python/FastAPI AI microservice platform to orchestrate PyTorch models for medical image analysis, enabling pathologists to use natural language for complex segmentation and classification tasks.
- Implemented an event-driven DAG workflow engine for async task execution, processing 10,000+ pathology images daily and streaming 12M+ imaging events through WebSockets with sub-10ms latency.
- Integrated generative and discriminative deep learning models into a unified service layer with asynchronous REST APIs for cross-department usage.

Software Engineering Intern | Information Technology of CAS Apr 2024 - Aug 2024 | Remote

- Designed Kafka and Redis-backed event streaming services that reduced p95 API latency from 2s to 200ms while processing millions of IoT events.
- Contributed to distributed Spring Boot monitoring services, delivered 35+ REST APIs, and maintained 99.7% uptime in Unix-based production environments.

OPEN SOURCE

nanoRL | github.com/RiddleHe/nanochat

- Led an open-source RL training framework inside nanochat for clean objective definitions, rollout experiments, RL algorithm research, reproducible ablations, and production-grade vLLM inference serving.

nanochat | github.com/RiddleHe/nanochat

- Built a hackable pretraining and chat stack supporting architecture definition and FLOP-controlled model ablations.

llm-interp | github.com/RiddleHe/llm-interp

- Authored reproducible interpretability scripts for model circuit research, including attention-sink analysis and reproductions of LLM decode indeterminism findings.

RESEARCH

Do Value Vectors in Deep Layers Need Context from the Residual Stream?

EMNLP 2026, under review

Co-first author. Challenged the standard attention mechanism of computing value vectors from the residual stream, discovering that deep layers benefit from context-free value vectors that preserve original token information and decouple optimal inputs for keys vs. values.

Proposed Bank of Values, a learned value-vector table for the last third of layers that eliminates the V cache and improves validation loss and average scores across 21 benchmarks on 135M and 780M models. arXiv:2606.02780

Vision Language Models Cannot Plan, but Can They Formalize?

ECCV 2026, under review

Studied whether VLMs can formalize visual planning tasks into solver-executable PDDL problem files. arXiv:2509.21576

A co-evolving agentic AI system for medical imaging analysis

Nature, under review

Built agentic medical-imaging workflows for tool selection, planning, knowledge updates, and human-in-the-loop co-evolution. arXiv:2509.20279

SKILLS

Languages: Python, Kotlin, Java, C++, JavaScript; ML/Infra: PyTorch, vLLM, Spark, Kafka, Iceberg, Delta Lake, FastAPI, Redis, Docker, Kubernetes, AWS, Spring Boot